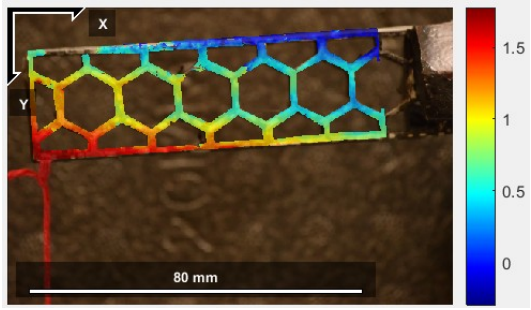
DIC analysis

Optical measurement of a trabecular specimen

MATLAB / Ncorr / Digital Image Correlation | Academic Year 2025/2026



Displacement field obtained by processing images using Ncorr.

DIC 2D

non-contact optical measurement using MATLAB/Ncorr

OBJECTIVE

Analyse the response of a 3D-printed nylon trabecular specimen using image correlation and compare its behaviour before and after damage.

MY CONTRIBUTION

I participated in the team's joint work, using brainstorming and collaboration across the experimental work, processing and discussion of results. Activities were shared rather than assigned exclusively to individual members.

ENGINEERING APPROACH

- Speckle preparation and image acquisition under loads applied using calibrated masses.
- ROI selection, subset and seed definition, scale calibration and field processing using Ncorr in MATLAB.
- Comparison of intact and damaged specimens and assessment of the influence of optical quality and correlation parameters.

RESULTS

The displacement fields are consistent with the expected macroscopic behaviour. The comparison shows greater bending of the damaged specimen under the same load. The work highlights the importance of controlled image acquisition for measurement quality.

PROJECT TEAM

Davide Calvelli / Giacomo Mommi / Francesco Paolucci